MTH 232(B) Dr. James Ashe
Final Exam Wednesday 15th May, 2013 **Name:**

No credit will be given unless you show your work.

1. The technique of integration used to find the indefinite integral $\int x^2 e^{-x}\,dx$ is

A. partial fractions **B.** by parts **C.** trigonometric substitution **D.** power rule

2. If $x > 1$, and $y = \int_1^x \sqrt{1+\sqrt{t}}\,dt$, then $\dfrac{dy}{dx}$ is equal to

A. $1+x$ **B.** $2x\sqrt{1+x}$ **C.** $\sqrt{1+\sqrt{x}}$ **D.** $\sqrt{1+x^2}$

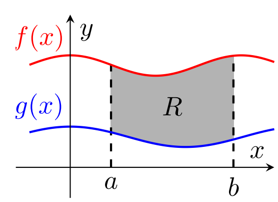
3. The *length of the arc* on the graph of $y = f(x)$ from the point $(a, f(a))$ to the point $(b, f(b))$ is computed using

A. $\int_a^b \sqrt{1+[f'(x)]^2}$ **B.** $\int_a^b \sqrt{1+[f(x)]^2}$ **C.** $\pi \int_a^b \sqrt{1+\dfrac{d^2y}{dx^2}}$ **D.** $\pi \int_a^b (f(x))^2\, x$

4. The integral expression which gives the area of the shaded region R in the figure is

A. $\int_a^b (g(x) - f(x))\; dx$ **C.** $\int_a^b (f(x) - g(x))^2\; dx$

B. $\pi \int_a^b (f(x) - g(x))\; dx$ **D.** $\int_a^b (f(x) - g(x))\; dx$



5. Let $f(x)$ be continuous on $[-5, 5]$ and $\int_0^5 f(x)\,dx = 13$. If for each x in $[-5, 5]$, $f(-x) = f(x)$, i.e., $f(x)$ is an even function, then $\int_{-5}^5 f(x)\,dx =$

A. 26 **B.** 13 **C.** 0 **D.** -13

6. Let $f(x)$ be continuous on $[-5, 5]$ and $\int_{-5}^0 f(x)\,dx = 13$. If for each x in $[-5, 5]$, $f(-x) = -f(x)$, i.e., $f(x)$ is an odd function, then $\int_{-5}^5 f(x)\,dx =$

A. 26 **B.** 13 **C.** 0 **D.** -13

7. The average value of a function f on the interval $[a, b]$ is given by

A. $\dfrac{1}{b-a}\int_a^b f(x)\,dx$ **B.** $(b-a)\int_a^b f(x)\,dx$ **C.** $\dfrac{b}{a}\int_a^b f(x)\,dx$ **D.** $\dfrac{f(b)-f(a)}{b-a}$

8. If $\int_1^2 f(x)\,dx = 5$, then $\int_1^2 (f(x) + 2x)\,dx$ is equal to

A. 5 **B.** 8 **C.** 10 **D.** $2x \cdot f(x)$

9. If $\int_b^a f(x)\,dx = 13$, then $\int_a^b f(x)\,dx =$

A. 13 **B.** 6.5 **C.** -13 **D.** 0

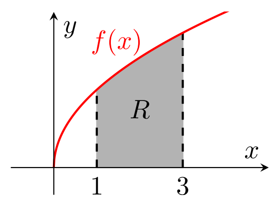
10. The volume of the solid generated by revolving the shaded region R about the x-axis is computed using the formula

A. $\pi \int_1^3 (f(x))^2\,dx$

C. $\int f(x)\,dx$

B. $\pi \int_1^3 f(x)\,dx$

D. $\pi \int_1^3 \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$



11. $\int_3^3 \frac{x^4 + x^2 + 1}{\sqrt{x^8 + x^4 + 2}}$

A. 9 **B.** 0 **C.** 6 **D.** $\frac{1}{\sqrt{3}}$

Find each indefinite integral for problems 12 through 16. *Show your work.*

12. $\int \frac{2x}{10 + x^2}\,dx$

13. $\int 3xe^{-x}\,dx$

14. $\displaystyle\int \frac{dx}{\sqrt{4-x^2}}$

15. $\displaystyle\int \frac{dx}{(x-2)(x+4)}$

16. $\displaystyle\int \cos^3 x \sin^2\ dx$

17. Find the area of the region bounded by the graphs of $y = x^2 + 2$ and $y = x - 2$ from $x = -1$ to $x = 1$.

18. Find the volume of the solid generated by revolving the region bounded by the the x-axis and the curve $y = \sqrt{x+1}$ from $x = 0$ to $x = 3$ around the x-axis.

19. Find the length of the arc on the graph of $y = \frac{2}{3}x^{3/2} + 9$ from the point $(0, 9)$ to the point $(3, 2\sqrt{3} + 9)$.

20. If $F(x)$ is an antiderivative of $f(x)$ on the interval $[a, b]$ and $F(a) = 2$, $F(b) = 5$ then $\int_a^b f(x)\,dx =$

A. 7 **B.** 5 **C.** 2 **D.** 3

21. Solve the initial value problem, $\frac{dy}{dx} = (x+2)^{1/2}$ and $y(0) = 1$.

22. Evaluate $\int_{-1}^{3} |x - 2|\,dx$.

23. Use the figure to evaluate $\int_{-4}^{4} f(x)\,dx$.